

Sign Language Detection using Machine Learning

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Autumn Semester 2023

1 Abstract

One of the major goals of machine learning is to make machines learn how to get things done in a way similar to humans, so it's pretty valid that understanding and generating language be one of the major concerns of machine learning. Sign language is the major source of communication of deaf people. Communication in sign language mainly happens via hand gestures and facial expressions. Building a robust model for sign language recognition will be very contributing in increasing the inclusivity of the society in general. This project aims to train a model to recognize sign language using Computer Vision and Deep Learning. The significance of this project lies in its potential to bridge the communication gap for individuals with hearing impairments by enabling real-time interpretation of sign language gestures through advanced machine learning algorithms.

2 Introduction

All learning algorithms can be broadly classified into two categories: Supervised Learning and Unsupervised Learning. This project leverages the power of Supervised Learning to decode the intricacies of sign language gestures. By using openCV for image processing and TensorFlow and Keras for Deep learning, this project tries to build a robust model for sign language recognition.

OpenCV (Open Source Computer Vision Library) is an open source computer vision and machine learning software library. The library has more than 2500 optimized algorithms, which includes a comprehensive set of both classic and state-of-the-art computer vision and machine learning algorithms. These algorithms can be used to detect and recognize faces, identify objects, classify human actions in videos, track camera movements, track moving objects, etc.

TensorFlow is an end-to-end open-source deep learning framework. TensorFlow is a symbolic math li-

brary used for neural networks and is best suited for dataflow programming across a range of tasks. It offers multiple abstraction levels for building and training models.

Keras is an effective high-level neural network Application Programming Interface (API) written in Python. This open-source neural network library is designed to provide fast experimentation with deep neural networks, and it can run on top of TensorFlow.

Throughout this report, we use the tools mentioned above to train our model as accurately as possible and at the same time avoid overfitting.

3 Past research on the topic

The initial investigations into sign language detection required users to utilize specialized wired "data-gloves," which, unfortunately, were prohibitively expensive. Furthermore, these systems predominantly centered on finger spelling, necessitating users to spell out individual letters to convey words. Subsequent research endeavors adopted a more practical approach. Tamura and Kawasaki, for instance, presented an early image processing system capable of recognizing 20 Japanese signs by matching cheremes. The progression of research in this domain witnessed a shift towards the adoption of Hidden Markov Models (HMMs), proving more efficient even when dealing with sparse datasets.

As Deep Learning methodologies gained prominence, subsequent investigations began incorporating Convolutional Neural Networks (CNNs) and Neural Networks (NNs) to discern hand gestures and facial expressions. A notable advancement in this trajectory is the DeepSign model, demonstrating an impressive 97 percent accuracy in recognizing Indian Sign Language.

The current landscape is characterized by extensive ongoing research, underscoring the commitment to refining and advancing sign language detection

methodologies. Noteworthy is the model’s capacity to achieve such results with minimal training data, showcasing the efficacy of contemporary techniques.

4 Creating a Dataset

In this project, meticulous consideration of various intricacies in dataset creation is paramount. One often overlooked aspect in the input data pertains to the background environment. Ensuring that diverse or subtly perturbed backgrounds do not unduly impact the predictive capabilities of our model is of utmost importance.

To address this concern, we propose a methodology that involves reserving the initial n frames captured exclusively for the development of an aggregated background baseline. These initial frames serve as a foundation for the actual input frames containing sign language signals. The rationale behind this approach is to mitigate the influence of varying backgrounds on the model’s predictions.

Consider, for example, a scenario where the environmental conditions include flickering light. In such cases, relying solely on the first frame as the background (which may capture dim lighting conditions) could lead to inaccuracies in predictions when subsequent frames with sign language signals exhibit bright light. To circumvent such discrepancies, we advocate for the aggregation of frames to form a more comprehensive background representation.

As each new frame is added within the initial set of n frames, the standard background undergoes modification. This dynamic approach enables the model to adapt to changing environmental conditions, ultimately enhancing its robustness to variations in lighting and background settings. The aggregate background serves as a more nuanced and adaptable reference point for the model, contributing to more accurate predictions in scenarios characterized by diverse or evolving backgrounds.

By adopting this strategy, we aim to bolster the resilience of the model to the challenges posed by dynamic backgrounds, thereby promoting consistent and reliable performance in real-world applications of sign language detection. We leverage OpenCV’s capabilities to create datasets. On the introduction of hand signs by users, OpenCV dynamically studies and identifies the outermost contours with the maximum area. This encompasses the hand sign. By harnessing OpenCV’s precision in background assessment and contour analysis, we ensure the dataset’s richness and accuracy, laying a foundation for effective sign language recognition and interpretation

through advanced machine learning algorithms. In pursuit of developing a robust and nuanced sign language recognition model, our team deliberately opted to construct a proprietary dataset rather than relying on existing repositories available on platforms such as GitHub. The decision to undertake this endeavor was primarily rooted in a strategic educational objective, aiming to gain a profound understanding of the intricacies associated with dataset creation and manipulation.

Our team diligently generated approximately 10,000 images, with team members assuming the role of models for the sign poses. This methodological approach not only fostered a hands-on learning experience but also facilitated an in-depth exploration of various aspects inherent to sign language recognition. By capturing our own dataset, we were able to delve into the complexities of background verification, region of interest identification, image thresholding, and contour analysis, thus elevating our comprehension of these crucial components in the development of a sign language recognition system.

The dataset was meticulously divided into a ratio of 70:30. This partitioning strategy ensures that our model is trained on a substantial portion of the dataset, while a distinct and unbiased subset is reserved for evaluating the model’s performance.

To ensure a fair and rigorous evaluation of our model, the images constituting the training set were exclusively derived from signs executed by four team members. Subsequently, one team member’s sign images were segregated and reserved as the test set. This meticulous allocation not only upholds the integrity of the evaluation process but also ensures that the model is subjected to a diverse range of signing styles and nuances during the testing phase. Consequently, this comprehensive methodology contributes to the development of a sign language recognition model that is both robust and proficient in interpreting a variety of signing expressions.



Original image of sign of digit "2"

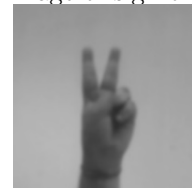


Image after Gaussian blurring



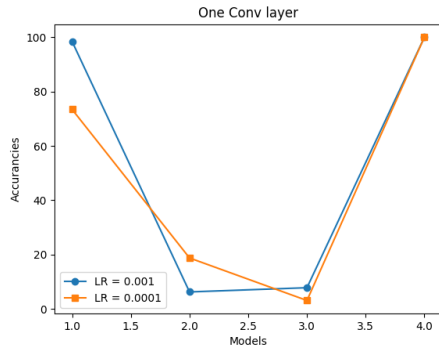
Thresholded image focusing only on sign

5 Results

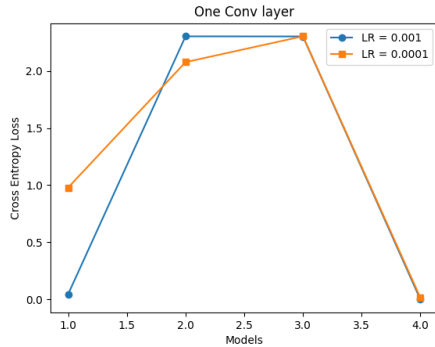
We trained 10 different models in total with tuning of hyper parameters such as learning rate and different number of convolutional layers, subsequent feed-forward layers and dropout probabilities.

List of models with **1** convolutional layer:-

- Model 1:- Dense layer = **64** neurons.
- Model 2:- Dense layers = **64, 128** neurons, dropout = **0.2**.
- Model 3:- Dense layers = **64, 128, 128** neurons, dropout = **0.2, 0.3**.
- Model 4:-Dense layers = **64, 128, 128** neurons, dropout = **None**.



Accuracy comparison for learning rates 0.001 and 0.0001 with 1 convolutional layer

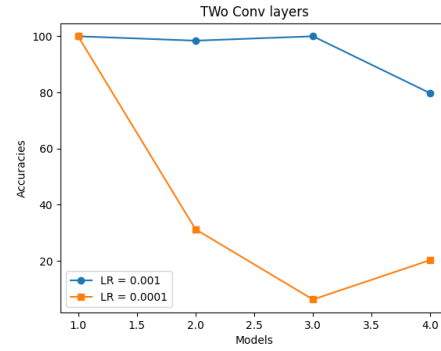


Cross entropy loss comparison for learning rates 0.001 and 0.0001 with 1 convolutional layer

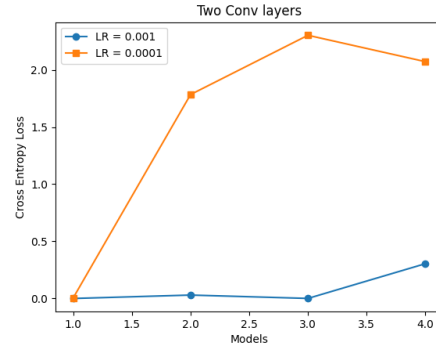
Best model :- Model **4**, dense layer with 64, 128, 128 neurons and no dropout ratio.

List of models with **2** convolutional layers:-

- Model 1:- Dense layer = **128** neurons.
- Model 2:- Dense layers = **64, 128** neurons, dropout = **0.2**.
- Model 3:- Dense layers = **64, 128** neurons, dropout = **0.2, 0.3**.
- Model 4:-Dense layers = **64, 128, 128** neurons, dropout = **0.2, 0.3**.



Accuracy comparison for learning rates 0.001 and 0.0001 with 2 convolutional layers

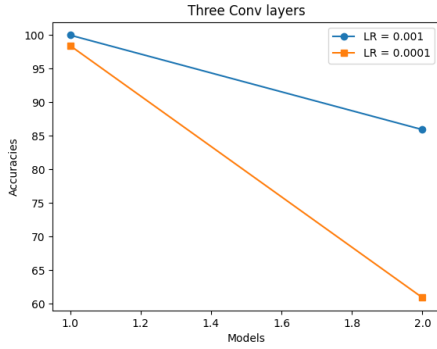


Cross entropy loss comparison for learning rates 0.001 and 0.0001

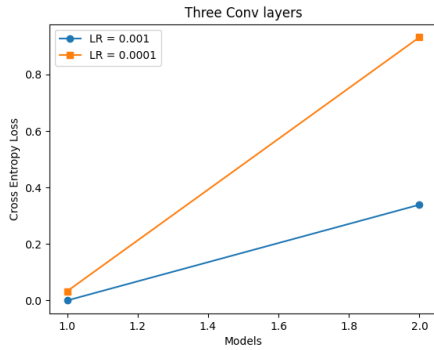
Best model :- Model **3**, dense layer with 64, 128 neurons, dropout ratios of 0.2, 0.3 and learning rate of 0.001.

List of models with **3** convolutional layers:-

- Model 1:- Dense layer = **64, 128** neurons, dropout ratio = **0.2**.
- Model 2:- Dense layers = **64, 128, 128** neurons, dropout = **0.2, 0.3, 0.3**.



Accuracy comparison comparison for learning rates 0.001 and 0.0001



Cross entropy loss comparison for learning rates 0.001 and 0.0001

Best model :- Model 1, dense layer with 64, 128 neurons, dropout ratios of 0.2 and learning rate of 0.001.

6 Future prospects

The current state of our model successfully predicts static sign language signals. However, a notable advancement entails extending its capabilities to encompass dynamic signs involving motion. The existing training paradigm involves singular frames with corresponding labels, mapping to specific static symbols. To accommodate the nuances of dynamic signals, a pivotal modification involves transitioning from singular frames to sequences of frames as training instances.

Incorporating sequential information into the training data enables the model to capture temporal dependencies inherent in dynamic sign language expressions. Unlike static signs, which can be adequately represented by isolated frames, dynamic signs unfold across a series of frames, requiring a more sophisticated neural network architecture.

To address this, the chosen neural networks must be adept at handling variable-sized inputs, as the introduction of frame sequences introduces a temporal dimension that demands a more nuanced approach than simple Convolutional Neural Networks (CNNs). Architectures such as Recurrent Neural Networks (RNNs) or Long Short-Term Memory Networks (LSTMs) are well-suited for processing sequential data.

RNNs, in particular, are designed to model temporal dependencies by maintaining internal states that capture information from previous time steps. LSTMs, as a variant of RNNs, are capable of learning long-range dependencies and are less susceptible to issues like the vanishing gradient problem.

By transitioning to a sequence-based training approach and leveraging neural networks designed for variable-sized inputs, our model can evolve to discern both static and dynamic sign language expressions. This adaptation is essential for capturing the temporal dynamics inherent in dynamic signs and will contribute to a more comprehensive and accurate sign language detection system.

Another potential enhancement for our model involves bolstering its capacity to detect multiple signals within a single frame, a capability that can be cultivated through the strategic implementation of data augmentation techniques. Integrating data augmentation methods becomes imperative for empowering the model with the ability to identify multiple signals within a singular frame. Techniques such as spatial transformations, random rotations, and scaling can diversify the dataset, exposing the model to a broader spectrum of scenarios and increasing its robustness in recognizing multiple signs concurrently.

7 Demonstration

Here is a link to the demonstration of the working of our model, and how it predicts the signals at an adept accuracy in real time. [DEMO](#).

8 References

- [DEEPSIGN: A DEEP-LEARNING ARCHITECTURE FOR SIGN LANGUAGE RECOGNITION](#) by Jai Shah
- [Visual Recognition of American Sign Language using Hidden Markov Models](#)
- [Blog link](#) The blog from which we got inspired and developed our fundamental convolutional neural networks.